Given: Apply the COCOMO II model to estimate the development effort and project cost. An additional recommendation feature increases the product size by 20%.

Part A: Estimate Development Effort and Project Cost

1. Assumed Parameters:

Parameter	Value
Initial software size	20 KLOC
Project complexity	High
Scale factor (B)	1.10
Effort multiplier (EAF)	1.20
Cost per person-month	₹50,000
A	2.94

2. COCOMO II Formula:

$$Effort = A \times (Size)^{E} \times EAF$$

Where,

$$E = B + 0.01 \times \sum SF$$

For simplicity, assume the combined exponent $E = 1.10$.

3. Calculate Effort:

$$Effort = 2.94 \times (20)^{1.10} \times 1.20$$

$$Effort = 2.94 \times 29.57 \times 1.20$$

\[\boxed{Effort=104.48\ Person\ Months} \]

4. Calculate Project Cost:

$$Cost = Effort \times Cost\,per\,Person\,Month$$

$$= 104.48 \times 50,000$$

\[\boxed{Cost=₹52,24,000} \]

Result (Part A):

- Estimated effort = 104.48 Person-Months.
- Estimated project cost = ₹52,24,000.

Part B: Re-estimate After Adding Recommendation Feature

1. Calculate New Software Size:

The product size increases by 20%.

$$New\ Size = 20 + \frac{20 \times 20}{100}$$

\[\boxed{New\ Size=24\ KLOC} \]

2. Calculate New Effort:

Using the same COCOMO II formula:

$$Effort = 2.94 \times (24)^{1.10} \times 1.20$$

$$Effort = 2.94 \times 32.18 \times 1.20$$

\[\boxed{Effort=113.60\ Person\ Months} \]

3. Calculate New Project Cost:

$$Cost = 113.60 \times 50,000$$

\[\boxed{Cost=₹56,80,000} \]

4. Calculate Increase in Cost:

$$Increase = 56,80,000 - 52,24,000$$

\[\boxed{Increase=₹4,56,000} \]

Percentage increase:

$$= \frac{4,56,000}{52,24,000} \times 100$$

\[\boxed{8.72\%} \]

Comparison of Results

Parameter	Part A	Part B
Software Size	20 KLOC	24 KLOC
Effort	104.48 PM	113.60 PM
Project Cost	₹52,24,000	₹56,80,000

Explanation

After adding the recommendation feature, the software size increases by 20%, from 20 KLOC to 24 KLOC. This increases the development effort from 104.48 to 113.60 person-months.

The project cost also increases from ₹52,24,000 to ₹56,80,000, which is an increase of ₹4,56,000 or approximately 8.72%.

Conclusion: COCOMO II helps estimate software development effort and cost. Adding new features increases software size, which results in higher development effort and project cost.